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# Auditing the Calibration of Eligibility Scores Produced by Large Language Models in Patient–Clinical Trial Matching

*Assessing the Effect of RLHF and Uncovering Mode Collapse in Open-Source Medical LLMs*

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# Abstract

**Keywords:** large language models, calibration, clinical trial matching, RLHF, medical AI, mode collapse, expected calibration error, conformal prediction

# Résumé

**Mots-clés :** grands modèles de langage, calibration, appariement d'essais cliniques, RLHF, IA médicale, mode collapse, expected calibration error, prédiction conforme

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# List of Abbreviations

| Acronym | Definition |
|---|---|
| AUC | Area Under the Curve |
| CI | Confidence Interval |
| ECE | Expected Calibration Error |
| EHR | Electronic Health Record |
| GPU | Graphics Processing Unit |
| IC | Intervalle de Confiance (French) |
| LLM | Large Language Model |
| LoRA | Low-Rank Adaptation |
| MC | Mode Collapse |
| ML4H | Machine Learning for Health |
| NEI | Not Enough Information |
| NF4 | Normal Float 4-bit (quantization) |
| NLL | Negative Log-Likelihood |
| NLP | Natural Language Processing |
| PPO | Proximal Policy Optimization |
| QA | Question Answering |
| RCT | Randomized Controlled Trial |
| RLHF | Reinforcement Learning from Human Feedback |
| SFT | Supervised Fine-Tuning |
| TREC | Text REtrieval Conference |

# Chapter 1

# Introduction

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**Table 2.1:** Comparative positioning of related works and this thesis.

| Work | Domain | Models | RLHF | Calib. | Recalib. |
|---|---|---|---|---|---|
| Jin et al. 2024 | Trial matching | GPT-4 | — | — | — |
| Kadavath 2022 | QA | Anthropic | ✓ | ✓ | — |
| Labrak et al. 2024 | Med QA | BioMistral | — | ✓ | — |
| **This work** | **Trial matching** | **6 LLMs** | ✓ | ✓ | ✓ |

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**Figure 4.1:** Overview of the experimental design: 6 models $\times$ 2 prompts $=$ 12 configurations evaluated on 1,015 patient-criterion pairs.

## 4.2 Research Question and Pre-registered Hypotheses

**Hypothesis 1** (H1). *Current LLM pipelines are poorly calibrated in clinical trial matching: ECE* $> 0.05$ *(threshold from Van Calster et al. 2019).*

**Hypothesis 2** (H2). *RLHF post-training degrades calibration:* $ECE(Instruct) > ECE(Base)$*, tested by permutation with Bonferroni correction.*

**Hypothesis 3** (H3). *LLM eligibility scores exhibit ordinal validity: C-index* $> 0.70$.

**Hypothesis 4** (H4). *Selective abstention improves clinical quality:* $F_1(80\%\ coverage) > F_1(100\%\ coverage) + 0.02$.

**Table 4.1:** Selected models and their scientific role.

| Model | Type | Scientific Role |
|---|---|---|
| Mistral-7B-v0.3 | Generalist, no RLHF | H2 control |
| Mistral-7B-Instruct-v0.3 | Generalist, with RLHF | H2 test |
| Llama-3.1-8B | Generalist, no RLHF | H2 replication |
| Llama-3.1-8B-Instruct | Generalist, with RLHF | H2 replication |
| Meditron-7B | Medical, no RLHF | Pure specialization effect |
| BioMistral-7B | Medical, inherited RLHF | Validation vs Labrak 2024 |

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$$\mathrm{ECE} = \sum_{k=1}^{K} \frac{|B_k|}{n} \left|\mathrm{acc}(B_k) - \mathrm{conf}(B_k)\right| \tag{4.1}$$

**Algorithm 1** Log-probability scoring for eligibility label prediction

**Require:** Patient note $n$, criterion $c$, criterion type $t$
**Ensure:** Predicted label $\hat{y}$, confidence $\hat{p}$, distribution $\pi$

Build prompt from $n$, $c$, $t$
$\mathcal{L} \leftarrow$ candidate labels according to $t$
**for all** $\ell \in \mathcal{L}$ **do**
  $s_\ell \leftarrow$ mean log-probability of tokens of $\ell$
**end for**
$\pi \leftarrow \mathrm{softmax}(s)$
$\hat{y} \leftarrow \arg\max_\ell \pi_\ell$
$\hat{p} \leftarrow \pi_{\hat{y}}$
**return** $\hat{y}, \hat{p}, \pi$

### 4.8.2 Brier Score

$$\text{Brier} = \frac{1}{n}\sum_{i=1}^{n}(p_i - y_i)^2 \tag{4.2}$$

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## 5.2 Pre-LLM Audit and Expert Agreement

**Finding 1** (F1: GPT-4's Selective Evasion Bias)**.** *GPT-4 over-uses the "not applicable" label (ratio 2.16 vs experts) but uses "not enough information" with a ratio of 0.99. This asymmetric pattern reveals a selective evasion, unreported in the original TrialGPT paper.*

## 5.3 Calibration Audit: H1 Supported Across All Models

**Table 5.1:** Calibration metrics across the 12 configurations.

| **Configuration** | **ECE** | **95% CI** | **Brier** |
|---|---|---|---|
| Mistral-Base A | 0.133 | [0.104, 0.163] | 0.258 |
| Mistral-Instruct A | 0.322 | [0.297, 0.350] | 0.311 |
| Mistral-Base B | 0.189 | [0.161, 0.218] | 0.269 |
| Mistral-Instruct B | 0.253 | [0.229, 0.287] | 0.353 |
| Llama-Base A | 0.124 | [0.100, 0.148] | 0.187 |
| Llama-Instruct A | 0.510 | [0.483, 0.535] | 0.456 |
| Llama-Base B | 0.151 | [0.123, 0.177] | 0.225 |
| Llama-Instruct B | 0.602 | [0.575, 0.628] | 0.558 |
| Meditron A | 0.182 | [0.153, 0.212] | 0.263 |
| Meditron B | 0.231 | [0.204, 0.259] | 0.283 |
| BioMistral A | 0.155 | [0.126, 0.182] | 0.234 |
| BioMistral B | 0.194 | [0.166, 0.222] | 0.244 |

**Figure 5.1:** Reliability diagrams of all 12 configurations (Prompt A left, Prompt B right). All curves fall well below the perfect calibration diagonal.

## 5.4 The Effect of RLHF: H2 Supported (Main Contribution)

**Finding 2** (F2: RLHF Systematically Degrades Calibration). *Across two independent model families (Mistral-7B and Llama-3.1-8B) and two independent prompt designs, RLHF-aligned Instruct versions exhibit significantly worse calibration than their Base counterparts (all* $p < 0.003$ *with Bonferroni correction,* $\Delta ECE$ *ranging from +0.064 to +0.451).*

## 5.5 The Enriched-Prompt Paradox

**Finding 3** (F3: The Enriched-Prompt Paradox). *Contrary to the intuition that expert prompt design improves calibration, prompts enriched with definitions, examples, and exclusion rules* degrade *calibration in 5 of 6 models. Only the strongly RLHF-aligned Mistral-Instruct benefits.*

## 5.6 Mode Collapse in Medical and RLHF-Aligned LLMs

**Finding 4** (F7: Mode Collapse in 5 of 12 Configurations). *Five open-source configurations — both BioMistral variants, all Llama-Instruct variants, and Llama-Base with prompt B — predict "not enough information" in more than 97% of cases. Their apparently reasonable ECE (0.15–0.60) reflects proximity to a trivial NEI baseline rather than genuine probabilistic quality.*

**Finding 5** (F8: Mode Collapse Is Structural, Not Superficial). *When forced to choose between the two decisive labels of each criterion type by removing evasion candidates from the softmax, collapsed models do not recover reasoning ability. Instead, collapse migrates from "not enough information" to the majority label per criterion type ("not excluded" for exclusion criteria, or "excluded" for one Llama-Base variant). Evasion behaviors mask absence of clinical reasoning rather than strategic uncertainty communication.*

## 5.7 Performance on Truly Discriminative Cases

**Finding 6** (F9: Global Accuracy Hides Complete Failure on Discriminative Cases). *On the 238 truly discriminative cases (23.4% of the dataset), Meditron achieves 0% accuracy despite its 64.8% global accuracy; BioMistral achieves 0.0% and 0.4% accuracy for its two prompts. The apparent superiority of medical LLMs disappears on the* *cases requiring genuine clinical decisions. Only Mistral-Instruct with prompt A reaches acceptable clinical performance (71.9% accuracy, ECE 0.054).*

**Table 5.2:** Global vs. non-trivial accuracy across configurations. The performance drop reveals which models rely on the majority class distribution.

| **Configuration** | **Global acc.** | **Non-trivial acc.** | **Drop** |
|---|---|---|---|
| Meditron A | 0.648 | 0.000 | −0.847 |
| Meditron B | 0.630 | 0.000 | −0.824 |
| BioMistral B | 0.289 | 0.000 | −0.377 |
| BioMistral A | 0.296 | 0.004 | −0.381 |
| Llama-Instruct B | 0.292 | 0.013 | −0.365 |
| Llama-Base B | 0.308 | 0.013 | −0.386 |
| Llama-Instruct A | 0.290 | 0.021 | −0.351 |
| Llama-Base A | 0.230 | 0.265 | +0.046 |
| Mistral-Base B | 0.586 | 0.416 | −0.222 |
| Mistral-Instruct B | 0.480 | 0.500 | +0.026 |
| Mistral-Base A | 0.500 | 0.613 | +0.149 |
| Mistral-Instruct A | 0.308 | **0.719** | +0.536 |

## 5.8 Systematic Inclusion vs Exclusion Asymmetry

**Finding 7** (F4: Inclusion vs Exclusion Asymmetry)**.** *LLM calibration on exclusion criteria is 2–3 times worse than on inclusion criteria across all generalist models.* MEDITRON *is the sole exception, showing the reverse pattern — likely due to its clinical guidelines training data.*

## 5.9 Model-Specific Evasion Biases

**Finding 8** (F5: Divergent Evasion Strategies Across LLMs)**.** *Whereas GPT-4 evades via the "not applicable" label (ratio 2.16), open- source LLMs consistently evade via "not enough information" (ratio 3.3 to 3.5 for BioMistral and Llama-Instruct). Evasion is systematic but channel-specific.*

**Figure 5.2:** Risk-coverage curves for the 6 Prompt-B configurations.

## 5.10 Ordinal Validity (H3) and Selective Classification (H4)

### 5.10.1 H3: Not Reached but Revealing Hierarchy

### 5.10.2 H4: Supported Only on the Mistral Family

## 5.11 Post-hoc Recalibration and Stability

**Finding 9** (F6: Post-hoc Isotonic Recalibration Works)**.** *Isotonic regression fitted on 70% of the data brings 11 of 12 configurations below the well-calibrated threshold of 0.05, robustly across 5-fold cross-validation. However, for collapsed models the "correction" merely calibrates a near-constant prediction and does not restore classification quality.*

## 5.12 Conformal Prediction and Inter-Model Consensus

### 5.12.1 Conformal Prediction Achieves Target Coverage

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## 6.2 Answer to the Research Question

## 6.3 Contributions

## 6.4 Limitations

### 6.4.1 External Validation

The present study relies on a single benchmark, TrialGPT-Criterion-Annotations [1], chosen because it is the only public corpus offering per-criterion multi-level expert annotations with an explicit inclusion/exclusion distinction — properties that are essential for fine-grained calibration analysis. External validation on independent datasets was not carried out, for three complementary reasons.

First, existing alternatives are not directly comparable. TREC Clinical Trials 2021/2022 and SIGIR 2016 operate at the trial level rather than the criterion level and use only three ordinal relevance grades, which precludes direct replication of the calibration protocol used here. Adapting the hypotheses to these datasets would not constitute a validation but a distinct study with different endpoints.

Second, the mode collapse phenomenon documented in Section 5.6 appears intrinsically linked to the availability of an evasion label ("not enough information") in the label space. Datasets that do not include such a label cannot be used to test whether this behavior generalizes: they would trivially prevent it by design.

Third, prospective validation on real clinical data would require ethical approval, hospital partnership, and integration into a live recruitment workflow. These constraints situate external validation as a research project in its own right, complementary to the present work rather than a subordinate step.

External validation nevertheless remains an important direction for future work.

### 6.4.2 Model Scale

### 6.4.3 Prompt Space

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### 6.5.3 Investigating the Origin of Mode Collapse

### 6.5.4 Extension to Larger and Multilingual Models

## 6.6 Closing Remarks

# Bibliography

[1] Q. Jin, Z. Wang, et al., “Matching patients to clinical trials with large language models”, *Nature Communications*, vol. 15, p. 9074, 2024.

# Appendix A

## Detailed Results Tables

# Appendix B

## Full Prompt Specifications

### B.1 Prompt A — Generic (Baseline)

```
Classify whether this patient meets the eligibility criterion.

Patient: {note}
Criterion ({type}): {criterion}

Choose exactly one label from: included, not included, excluded
    ,
not excluded, not enough information, not applicable.

Answer:
```

**Listing B.1:** Prompt A for Instruct models

### B.2 Prompt B — Adapted with Expert Guidance

# Appendix C

## Individual Reliability Diagrams

**(a)** Mistral-Base, Prompt A

**(b)** Mistral-Base, Prompt B

**Figure C.1:** Reliability diagrams for the 12 configurations.

# Appendix D

## Code and Data Availability

### D.1 Repository Structure

### D.2 Data Availability

### D.3 Reproducing the Experiments

### D.4 Computational Requirements

# Appendix E

## Generative AI Usage Declaration

### Tools Used

During the preparation of this thesis, the following generative AI tool was used as an assistant:

- **Claude** (Anthropic), primarily the Sonnet and Opus series model families, accessed via the `claude.ai` interface.

### Purpose of Use

Generative AI was used for:

- **Ideation and design discussions**: brainstorming experimental designs, discussing methodological trade-offs, and refining the research question.
- **Code assistance**: writing and debugging Python code for the evaluation pipeline, prompt engineering iterations, and statistical analyses.
- **Writing assistance**: refining the phrasing of prompts, drafting progress reports, and revising English formulations of methodological descriptions.
- **Bibliographic exploration**: identifying candidate references and papers relevant to the state of the art.

### Purpose of Non-Use

Generative AI was *not* used for:

- Generating experimental data or results.
- Fabricating statistical outcomes or numerical values.
- Producing scientific claims not backed by our own experiments.

- Writing the final thesis text verbatim (the author remains responsible for all formulations).

## Verification and Author Responsibility

All AI-generated content used in this work was:

- Reviewed, verified, and adapted by the author before inclusion.
- Cross-checked against primary sources for any factual claim.
- Understood by the author, who takes full responsibility for the scientific and methodological choices reflected in this thesis.

The scientific reasoning, experimental design, execution, and interpretation of results are the sole responsibility of the author.

# Appendix F

## Detailed Mode Collapse Analysis

### F.1 Prediction Distribution per Configuration

### F.2 Simulation: Forcing Decisive Predictions

### F.3 Composite Utility Score Ranking

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EOF echo "OK"